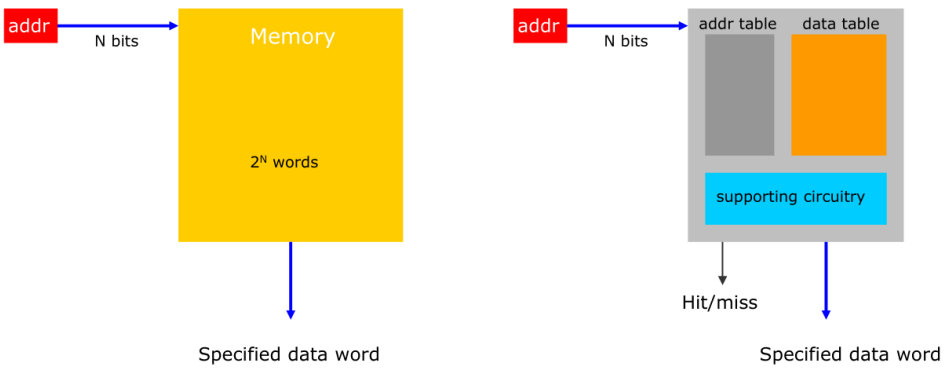
Cache memories: exercises

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1

Organization

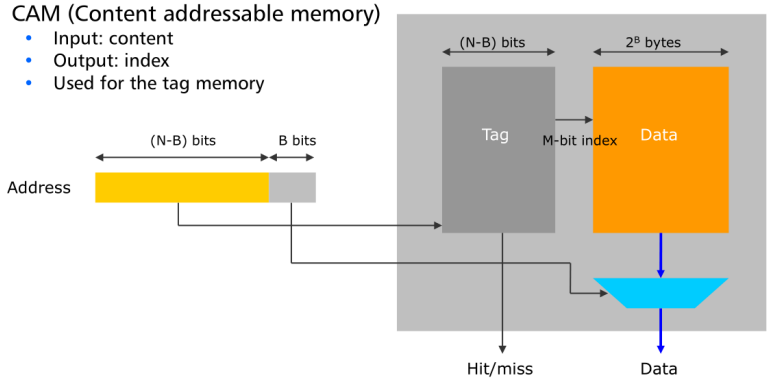


2

(Fully) Associative Mapping

- 2^B byte block
- 2^M entries
- N-bit address

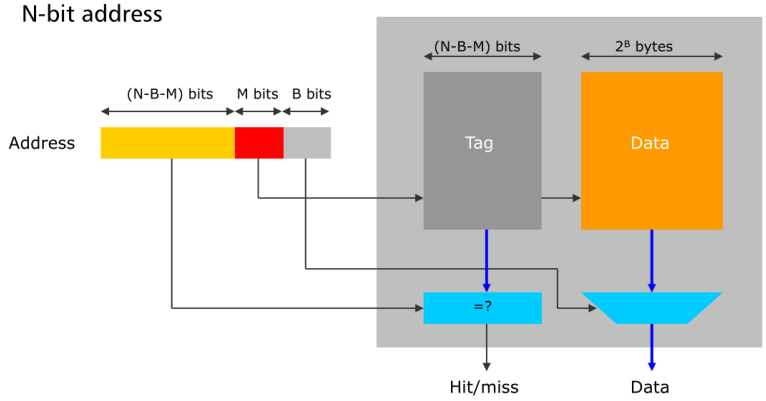
- CAM (Content addressable memory)
  - Input: content
  - Output: index
  - Used for the tag memory



3

Direct Mapping

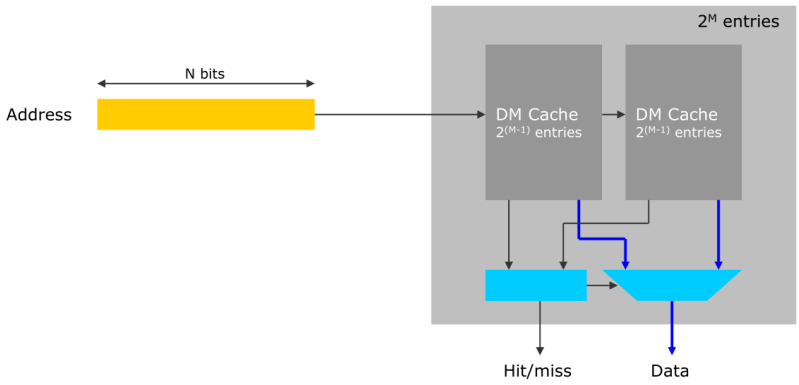
- 2^B byte block
- 2^M entries
- N-bit address



4

2-Way Set Associative Example

- 2^B byte block
- 2^M entries
- N-bit address



5

Exercise #1: Cache and memory characteristics

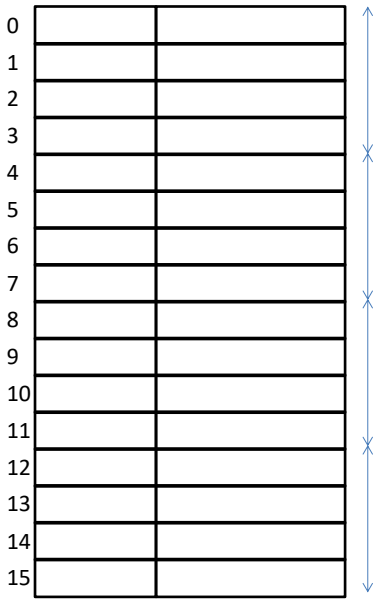
- Memory size: 2^{16} byte
- Cache organization
  - 16 lines
  - 1 line: 32 bytes
  - Mapping: 4-ways set associative
  - Replacement policy: LRU
- Initially
  - The cache is empty
  - Within each cache set, liness are filled according to their index

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Problem

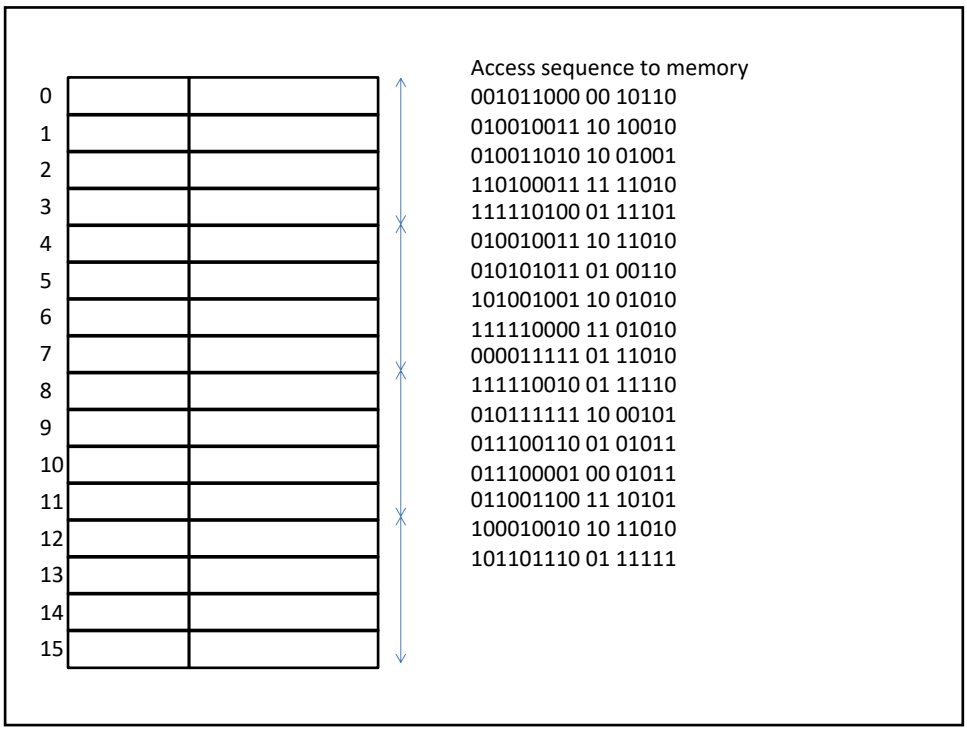
- Determine the block present on each cache line at the end of the following memory access sequence:

7

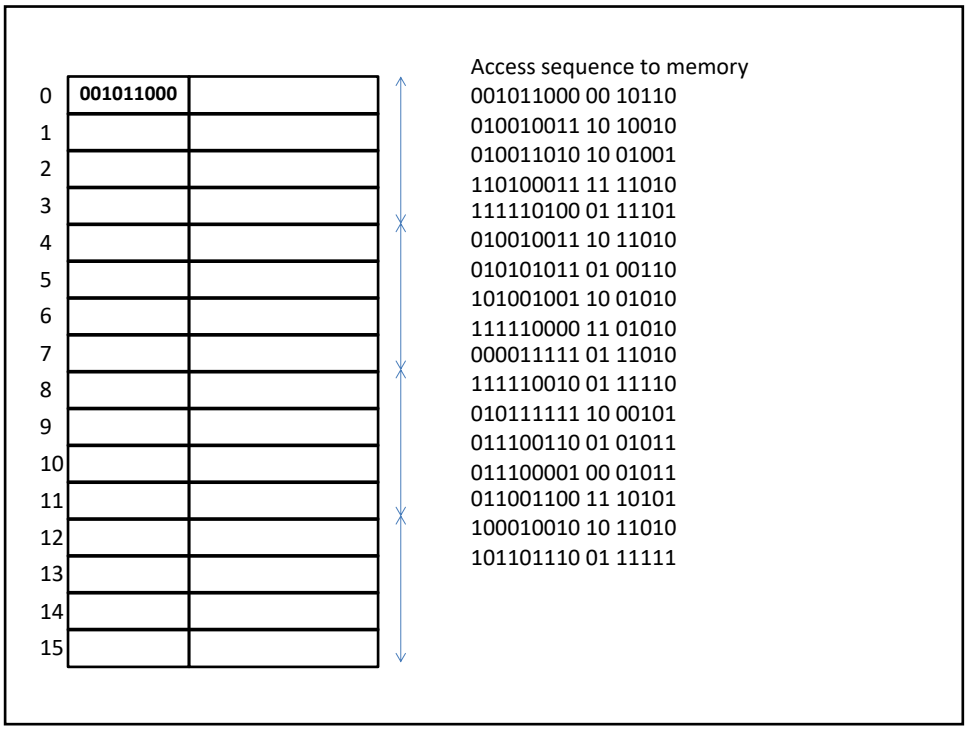


Access sequence to memory
0010110000010110
0100100111010010
0100110101001001
1101000111111010
1111101000111101
0100100111011010
0101010110100110
1010010011001010
1111100001101010
0000111110111010
1111100100111110
0101111111000101
0111001100101011
0111000010001011
0110011001110101
1000100101011010
1011011100111111

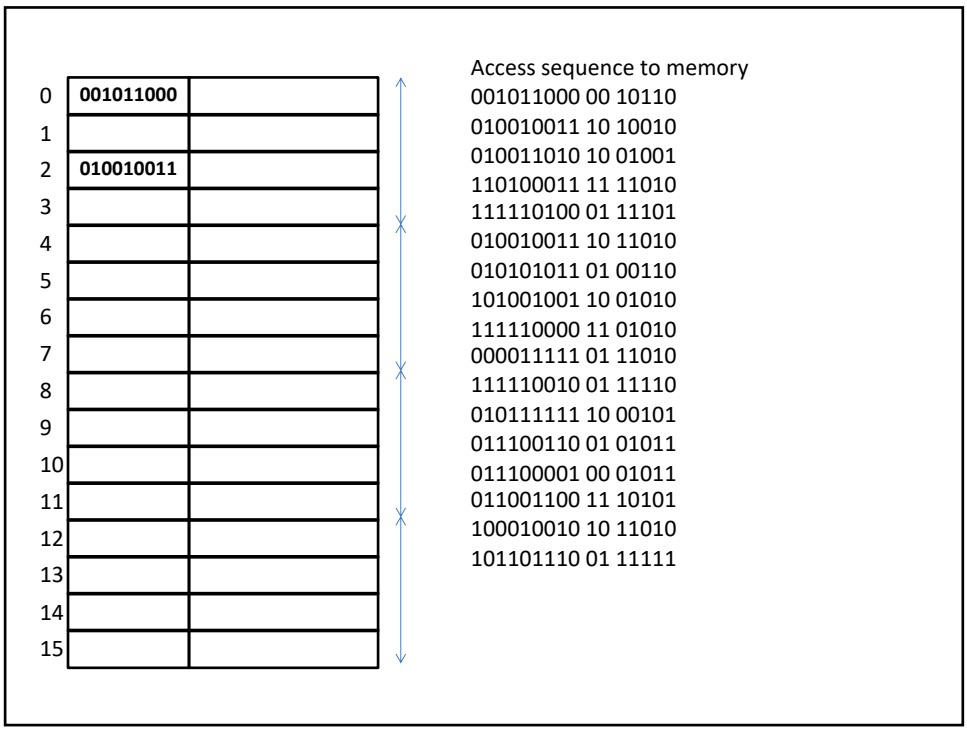
8


0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
Access sequence to memory
001011000 00 10110
010010011 10 10010
010011010 10 01001
110100011 11 11010
111110100 01 11101
010010011 10 11010
010101011 01 00110
101001001 10 01010
111110000 11 01010
000011111 01 11010
111110010 01 11110
010111111 10 00101
011100110 01 01011
011100001 00 01011
011001100 11 10101
100010010 10 11010
101101110 01 11111

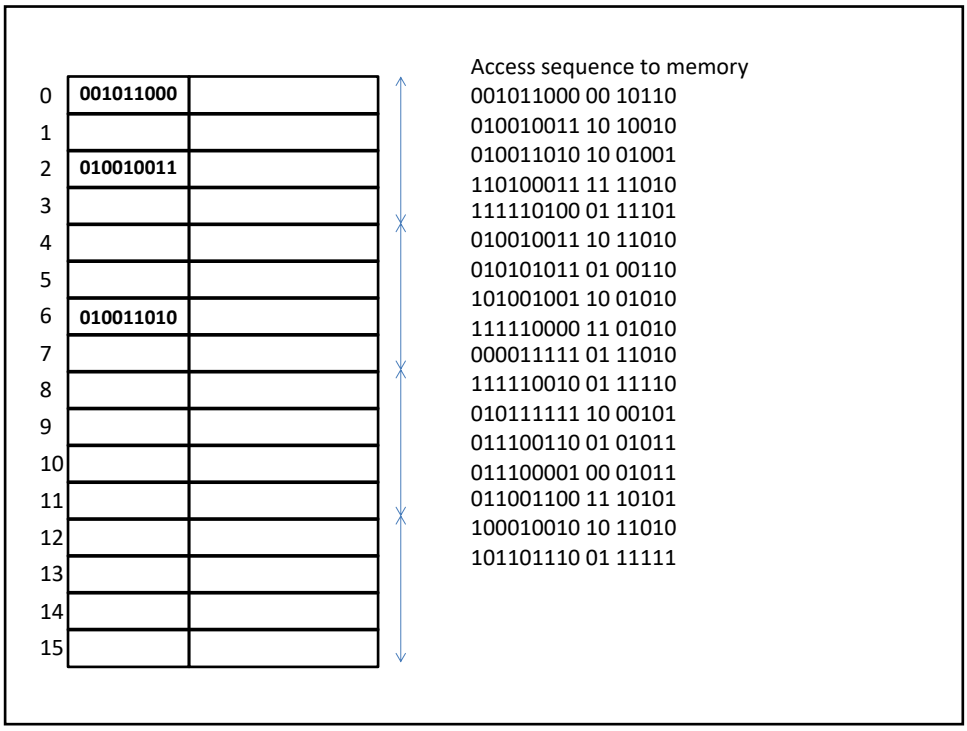
9


0
001011000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
Access sequence to memory
001011000 00 10110
010010011 10 10010
010011010 10 01001
110100011 11 11010
111110100 01 11101
010010011 10 11010
010101011 01 00110
101001001 10 01010
111110000 11 01010
000011111 01 11010
111110010 01 11110
010111111 10 00101
011100110 01 01011
011100001 00 01011
011001100 11 10101
100010010 10 11010
101101110 01 11111

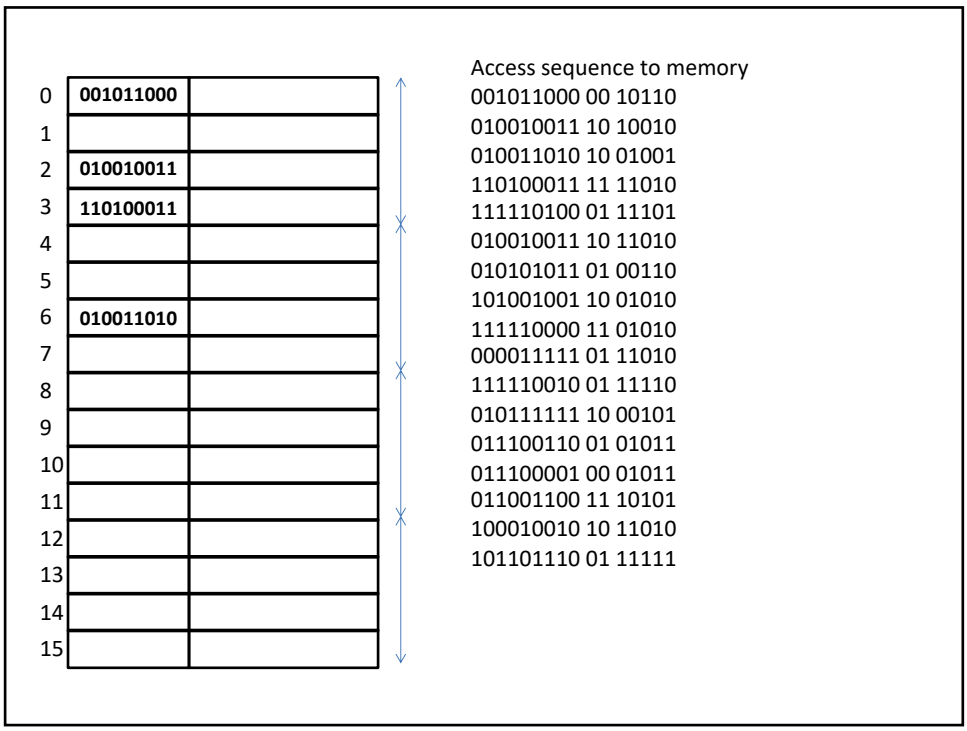
10


0 001011000
1
2 010010011
3
4
5
6
7
8
9
10
11
12
13
14
15
Access sequence to memory
001011000 00 10110
010010011 10 10010
010011010 10 01001
110100011 11 11010
111110100 01 11101
010010011 10 11010
010101011 01 00110
101001001 10 01010
111110000 11 01010
000011111 01 11010
111110010 01 11110
010111111 10 00101
011100110 01 01011
011100001 00 01011
011001100 11 10101
100010010 10 11010
101101110 01 11111

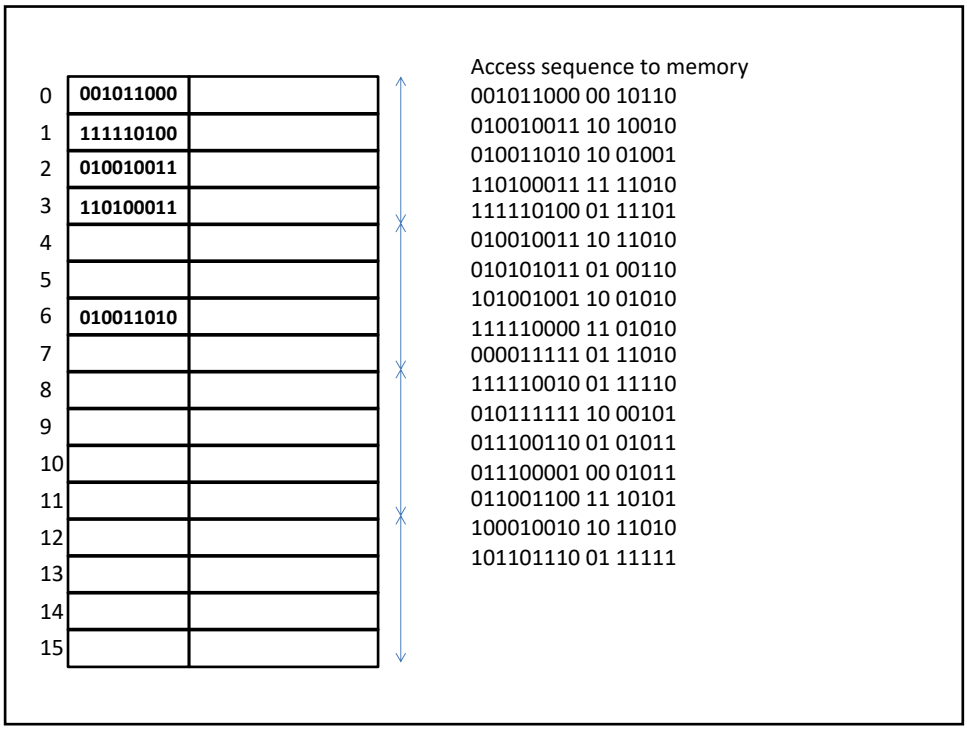
11


0 001011000
1
2 010010011
3
4
5
6 010011010
7
8
9
10
11
12
13
14
15
Access sequence to memory
001011000 00 10110
010010011 10 10010
010011010 10 01001
110100011 11 11010
111110100 01 11101
010010011 10 11010
010101011 01 00110
101001001 10 01010
111110000 11 01010
000011111 01 11010
111110010 01 11110
010111111 10 00101
011100110 01 01011
011100001 00 01011
011001100 11 10101
100010010 10 11010
101101110 01 11111

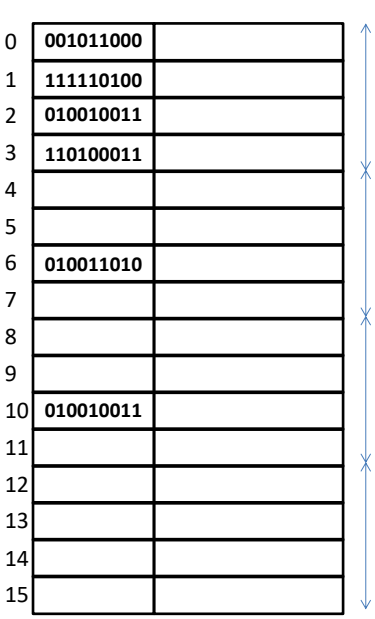
12


0 001011000
1
2 010010011
3 110100011
4
5
6 010011010
7
8
9
10
11
12
13
14
15
Access sequence to memory
001011000 00 10110
010010011 10 10010
010011010 10 01001
110100011 11 11010
111110100 01 11101
010010011 10 11010
010101011 01 00110
101001001 10 01010
111110000 11 01010
000011111 01 11010
111110010 01 11110
010111111 10 00101
011100110 01 01011
011100001 00 01011
011001100 11 10101
100010010 10 11010
101101110 01 11111

13


0 001011000
1 111110100
2 010010011
3 110100011
4
5
6 010011010
7
8
9
10
11
12
13
14
15
Access sequence to memory
001011000 00 10110
010010011 10 10010
010011010 10 01001
110100011 11 11010
111110100 01 11101
010010011 10 11010
010101011 01 00110
101001001 10 01010
111110000 11 01010
000011111 01 11010
111110010 01 11110
010111111 10 00101
011100110 01 01011
011100001 00 01011
011001100 11 10101
100010010 10 11010
101101110 01 11111

14



Access sequence to memory
001011000 00 10110
010010011 10 10010
010011010 10 01001
110100011 11 11010
111110100 01 11101
010010011 10 11010
010101011 01 00110
101001001 10 01010
111110000 11 01010
000011111 01 11010
111110010 01 11110
010111111 10 00101
011100110 01 01011
011100001 00 01011
011001100 11 10101
100010010 10 11010
101101110 01 11111

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Exercise #2:

- Let consider a system with the following characteristics:
  - direct mapped cache with 8 lines
  - memory composed of 64 blocks
- Let assume that the CPU acccessed in sequence to the following blocks: 21, 28, 30, 10, 7, 60, 30, 31, 32, 33, 60, 61, 25, 28, 27, 14.
- What is the content of each block of the cache?

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